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CONDITIONALS

BOOLEAN & CONDITIONALS

Start

BOOLEAN DATA TYPE

- Boolean (bool) is a data type that stores only two values:
  - true
  - false
- Used to represent yes/no, on/off, correct/incorrect conditions
- Commonly used in conditionals and loops

CONDITIONALS

- **Conditionals** allow a program to make decisions
- They execute different code depending on whether a condition is:
  - true
  - false
- Conditionals help programs behave intelligently

Real-life example:

- If it is raining → take an umbrella
- Otherwise → don’t take one

CONDITIONALS/COMPARISION CONDITIONS

Operator	Meaning	Example
==	Equal to	x == y
!=	Not equal to	x != y
>	Greater than	x > y
<	Less than	x < y
>=	Greater or equal	x >= y
<=	Less or equal	x <= y

CONDITIONALS/COMPARISION CONDITIONS

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C++ CONDITION STATEMENTS

- IF
- ELSE
- ELSE IF
- SWITCH

THE IF STATEMENT

- Use the if statement to specify a block of C++ code to be executed if a condition is true.

EXAMPLE:

```
int age = 18;

if (age >= 18) {
    cout << "You can vote.";
}
```

THE IF STATEMENT

DEMO

THE ELSE STATEMENT

- Use the else statement to specify a block of code to be executed if the condition is false.
- Provides an alternative path

THE ELSE STATEMENT

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THE ELSE IF STATEMENT

- Use the else if statement to specify a new condition to test if the first condition is false.
- You can use else if to check multiple conditions, one after another.

THE ELSE IF STATEMENT

DEMO

SUMMARY

- bool stores true or false
- Conditionals let programs make decisions
- Comparison operators are used to form conditions
- if runs code when a condition is true
- else runs when the condition is false
- else if checks multiple conditions
- Conditionals are essential for logic and control flow in C++

QUESTIONS?

ACTIVITY

Create a simple C++ program. that uses if & else.

1.create an int variable age with your age as the value.

2.create an if statement with the condition that if the age is above or equal 18 it should print “young adult”.

3. then an else statement that prints “minor”